Autonomous equations and equilibrium solutions

So far we've been focusing entirely on first order ordinary differential equations. We've looked at linear equations and nonlinear equations, but now we want to introduce autonomous equations as third type of ordinary differential equation.

Autonomous equations

Autonomous equations are differential equations in the form

$$\frac{dy}{dt} = f(y)$$

The defining characteristic of autonomous equations is that the independent variable never appears explicitly in the equation, but instead only as part of the derivative. In other words, in the autonomous equation above, the independent variable t only appears in dy/dt, not as part of $f(y)$, because f is a function only in y.

Technically, autonomous equations are separable equations that can be rewritten as

$$\frac{1}{f(y)}\, dy = dt$$

But our goal really won't be to solve them as separable equations. Instead, we're going to focus on a different kind of analysis. Autonomous equations are especially common in modeling real-world applications, so how we work with these equations will really be informed by the

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applications. We'll look at some of these applications later, but for now, let's turn our attention toward the direction fields of autonomous equations.

Direction fields

Looking again at the general form of the autonomous equation,

$$\frac{dy}{dt} = f(y)$$

we can see that, for any value y, we'll find one slope dy/dt. Because if we choose some value of y, let's call it y_0, we'll plug that into the right side of the autonomous equation to find $f(y_0)$, which will be some constant value, let's call it k. In other words,

$$f(y_0) = k$$

for some specific y_0. But if $f(y_0) = k$, then $dy/dt = k$. Therefore, for any particular y_0, the slope dy/dt will be constant.

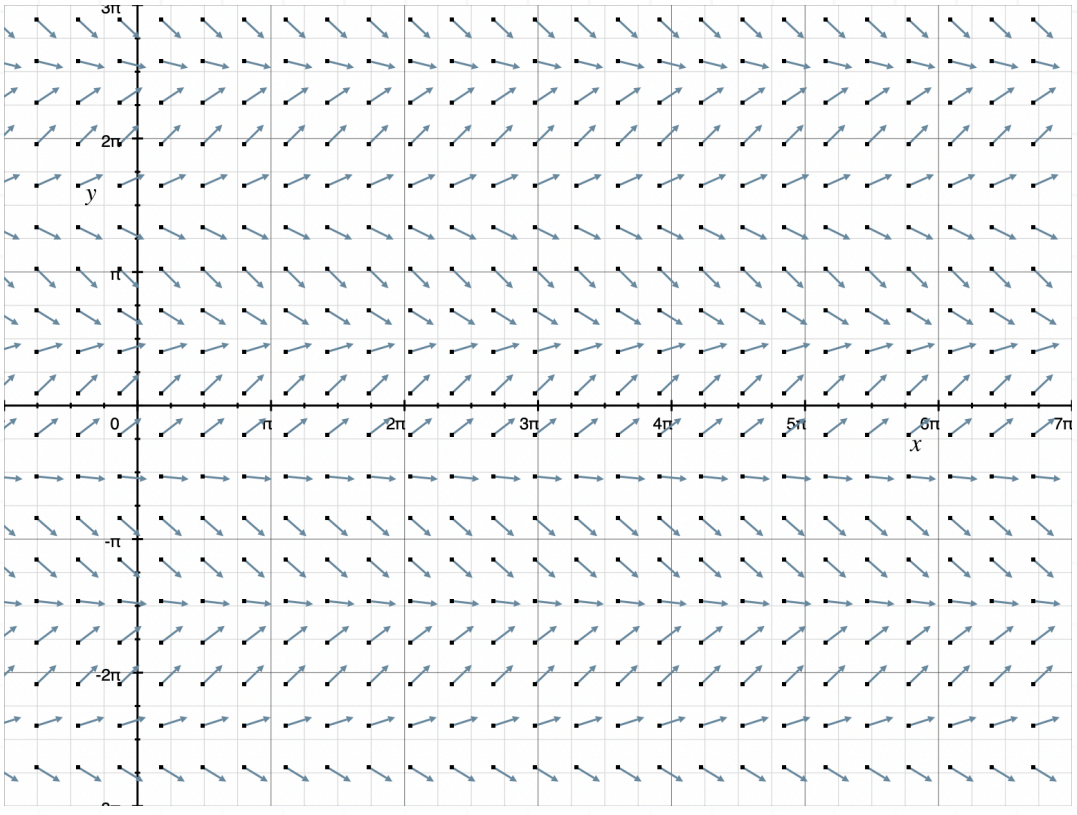
For example, consider the autonomous equation

$$\frac{dy}{dt} = \cos y$$

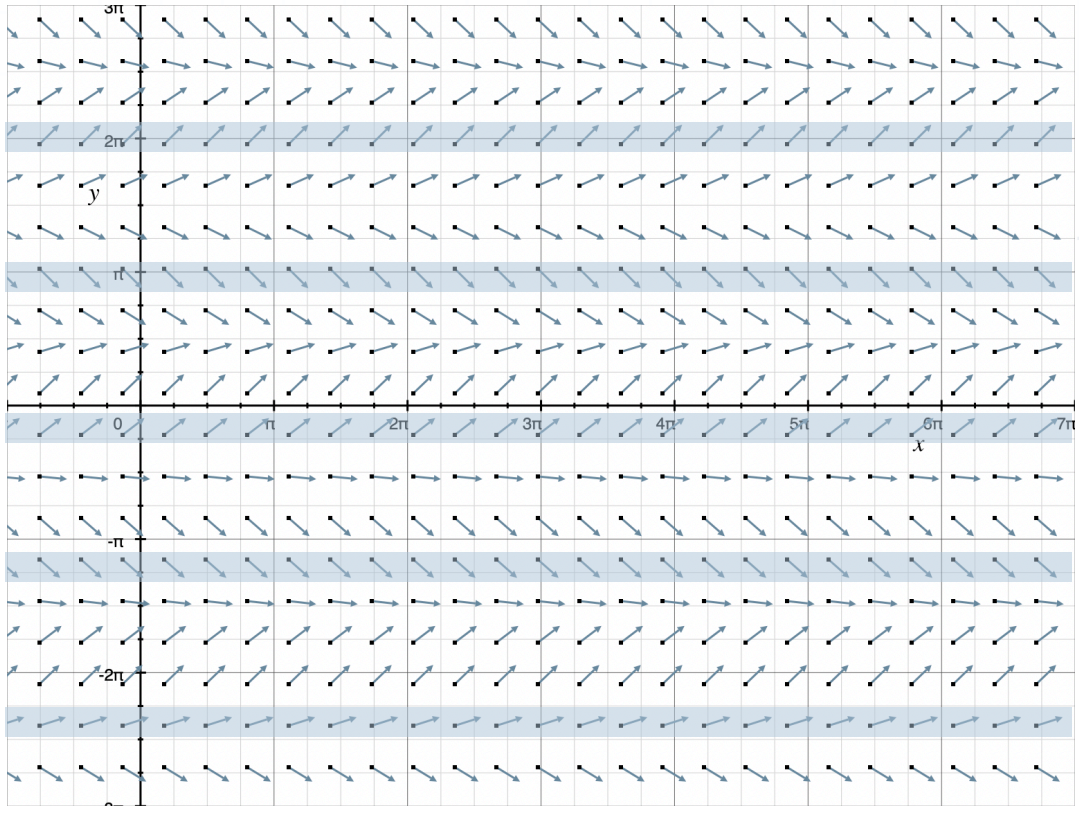
and its direction field

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Notice how every horizontal row contains a set of entirely parallel direction arrows. This will always be the case for the direction fields of autonomous equations.



Equilibrium solutions and stability

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We said earlier that we could pick some specific y_0, plug it into f from the right side of the autonomous differential equation, and come out with some constant value k, such that $f(y_0) = k$. If $k = 0$, it means that the function $f(y)$ has a zero at y_0.

We say that these zeros are the **critical points**, equilibrium points, or stationary points of the autonomous differential equation. In other words, a real number y_0 is a critical point of the autonomous equation if $f(y_0) = 0$.

Of course, when we get a zero value for the right side of the autonomous differential equation, we find the equation $dy/dt = 0$. This tells us that the slope dy/dt is 0, and when we have a zero slope, we have a perfectly horizontal line.

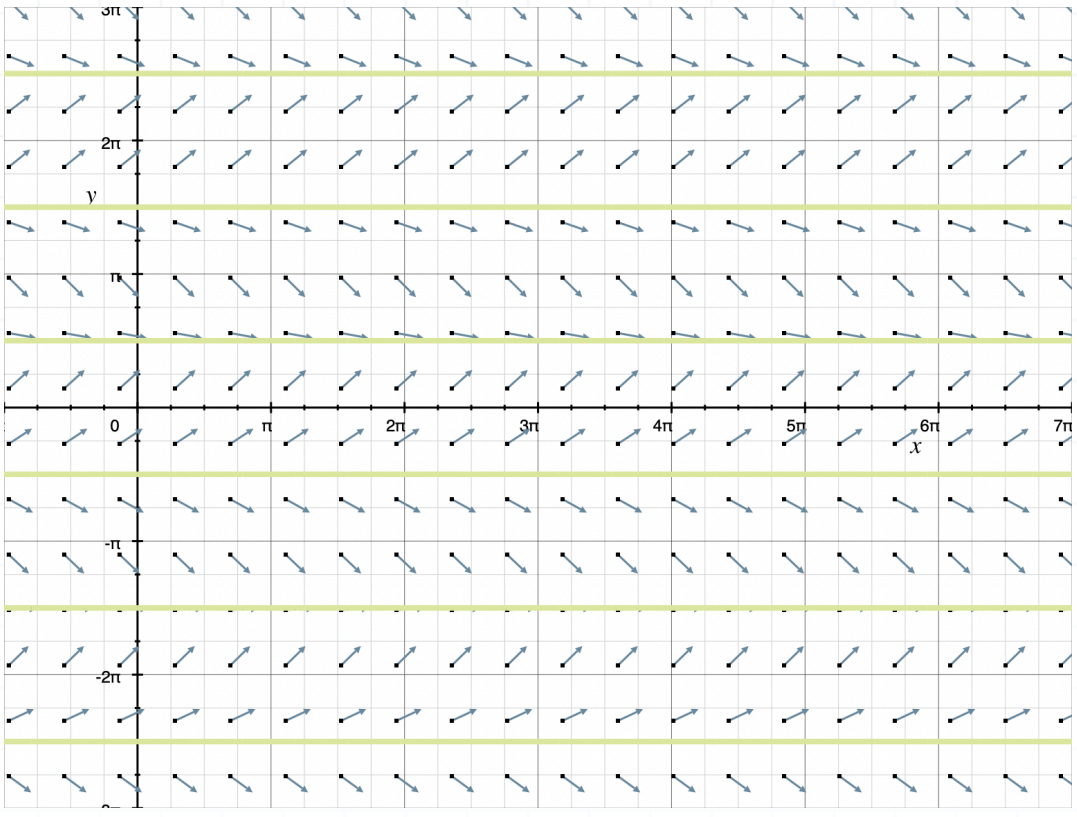
Bringing back the autonomous equation,

$$\frac{dy}{dt} = \cos y$$

we know $\cos y = 0$ when $y = (\pi/2) + \pi k$, where k is any integer. So in the direction field, we'll sketch in the **critical solutions**, or equilibrium solutions, which are the perfectly horizontal lines representing the solution curves that correspond to each critical point. Adding $y = \pm\,\pi/2,\ \pm\,3\pi/2$, and $\pm 5\pi/2$ to the direction field gives

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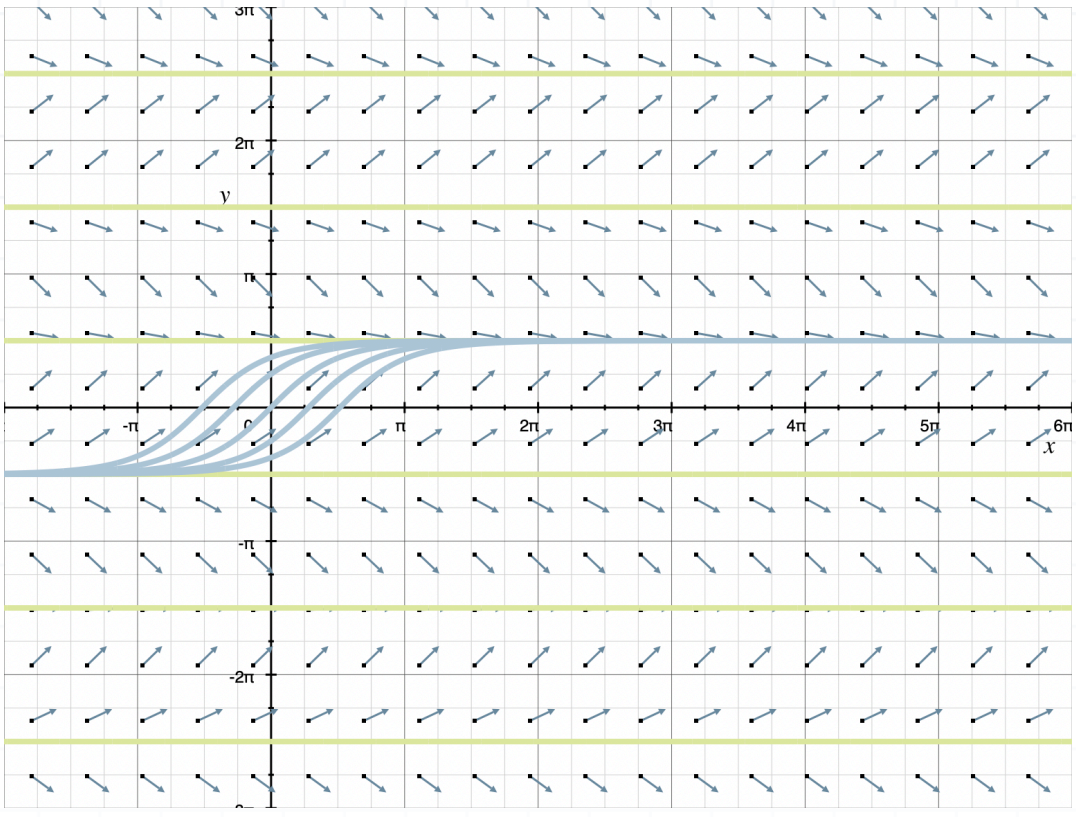




These equilibrium solutions are lines that will never be crossed by the curves of other solutions. Which means that, whenever we get close to an equilibrium solution, the slopes of the direction arrows level out.

To get a little more detailed, let's drill down to the part of the direction field that sits between $y = -\pi/2$ and $y = \pi/2$. If we sketch in some solution curves between those values, specifically the solutions through the initial conditions $y(0) = 0$, $y(0) = \pm\pi/4$, and $y(0) = \pm 3\pi/8$, we find these curves:

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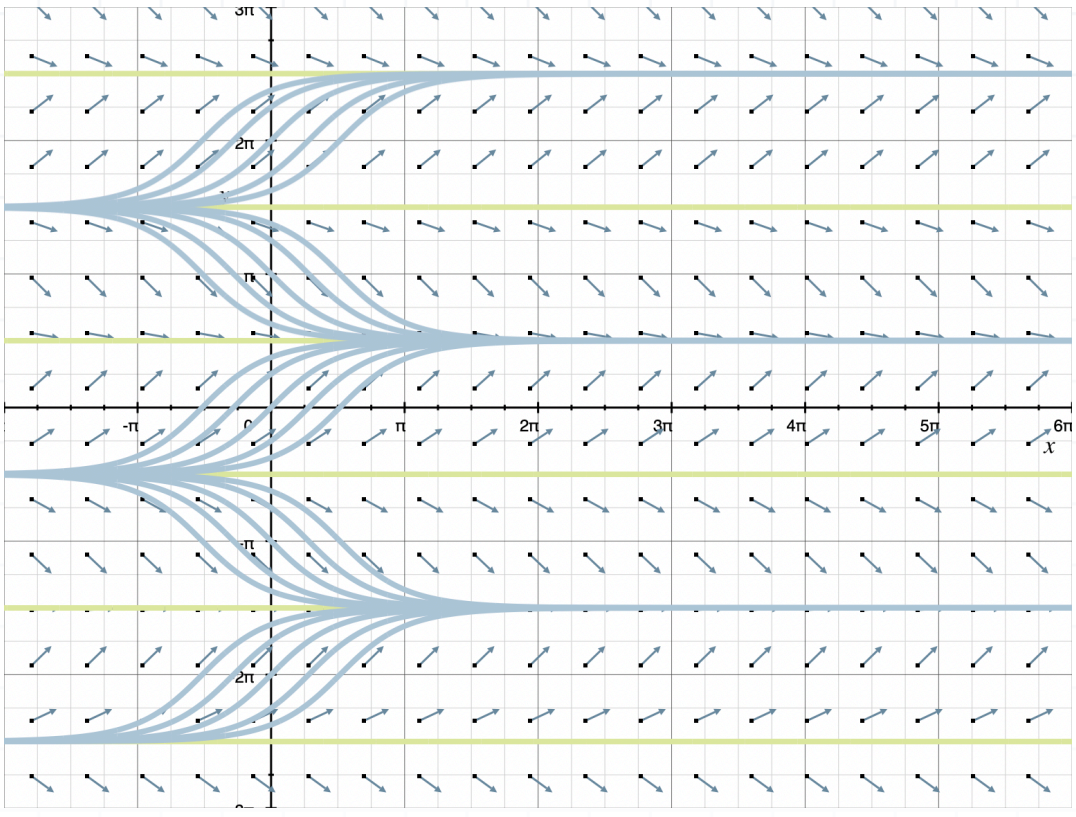


Notice how they all refuse to cross either of the equilibrium solutions at $y = \pm \pi/2$. What we can conclude is that the graph of any solution curve through some y_0 where $-\pi/2 < y_0 < \pi/2$, will remain inside the boundaries of $y = \pm \pi/2$.

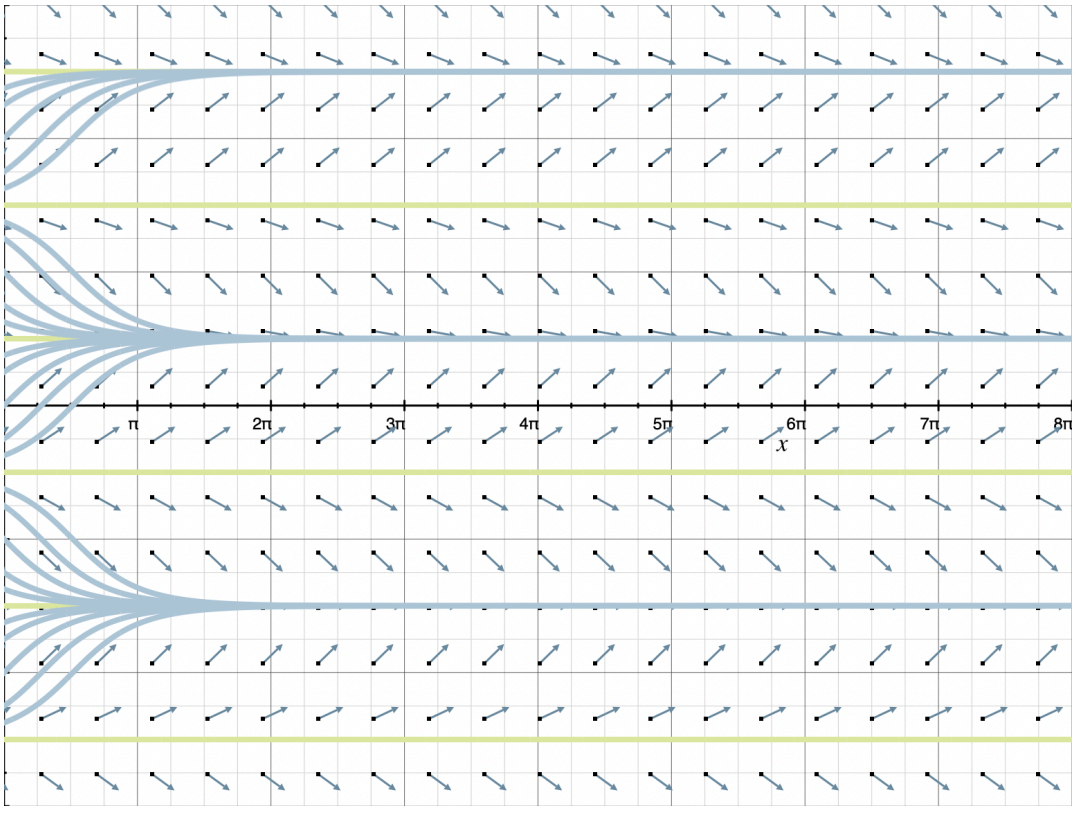
Furthermore, because of the fact we mentioned earlier that all the direction arrows along any horizontal line are parallel, all the solution curves within any region between two equilibrium solutions will be translations of each other. When we look at the five solution curves we sketched between $y = \pm \pi/2$, we notice that they all have the same shape, and they're just shifted left and right.

We find similar sets of shifted solution curves between the other equilibrium solutions.

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Now that we have a picture of several solution curves, we want to use them to classify the nature of each equilibrium solution. To do that, we should focus just on what's happening to the right of the vertical axis.



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What we see is that the solution curves are moving toward $y = -3\pi/2$, toward $y = \pi/2$, and toward $y = 5\pi/2$, while they're moving away from $y = -5\pi/2$, away from $y = -\pi/2$, and away from $y = 3\pi/2$.

Equilibrium solutions that attract solution curves are **stable equilibrium solutions**, or attractors, and equilibrium solutions that repel solution curves are **unstable equilibrium solutions**, or repellers.

Equilibrium solutions that attract solution curves on one side and repel them on the other are **semi-stable equilibrium solutions**. These solutions are usually more rare than stable and unstable solutions, and we don't have any of these in the direction field we've sketched for $dy/dt = \cos y$.

Let's do an example where we walk through how to find and classify the equilibrium solutions of an autonomous differential equation.

Example

Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = y^2 + 3y + 2$$

The autonomous differential equation has equilibrium solutions at

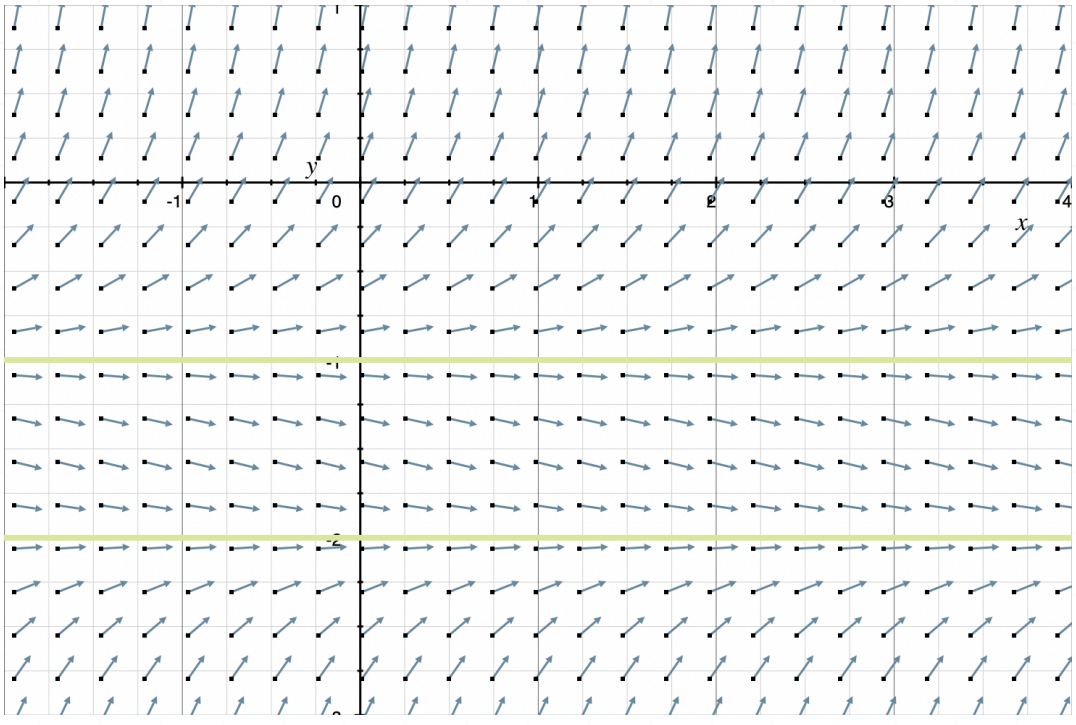
$$y^2 + 3y + 2 = 0$$

$$(y + 2)(y + 1) = 0$$

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$$y = -1, \ -2$$

If we sketch the direction field and then overlay the equilibrium solutions, we get



These two equilibrium solutions divide the vertical axis into three intervals:

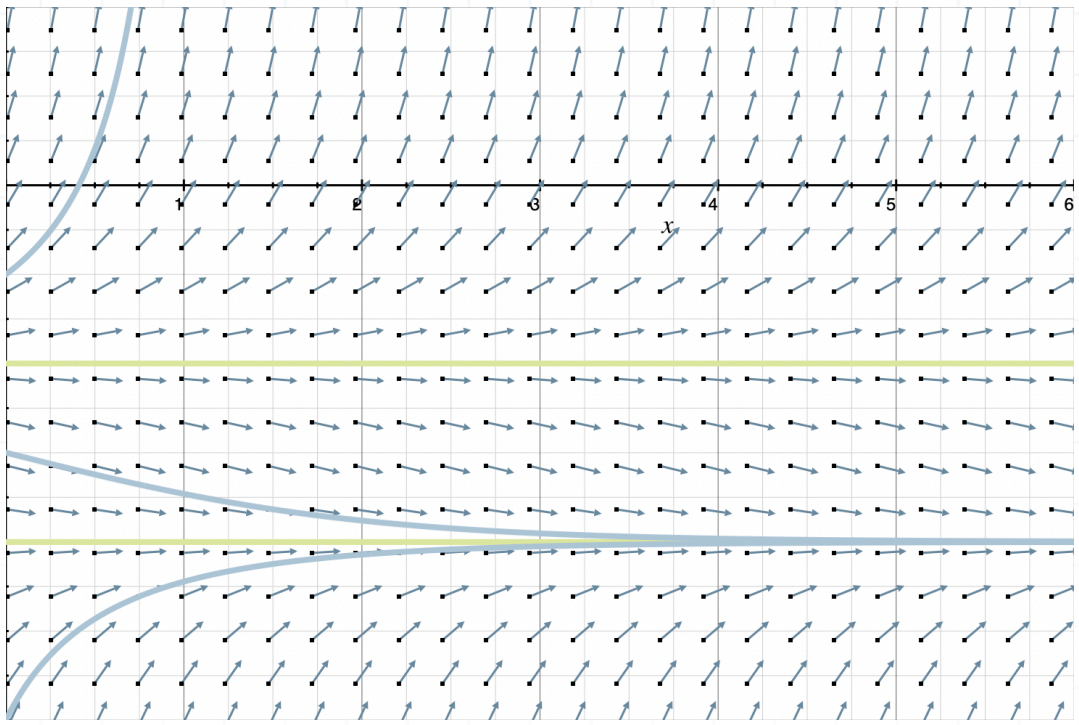
$$y < -2$$

$$-2 < y < -1$$

$$-1 < y$$

We can use the direction field to sketch a solution curve inside each of these intervals.

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These solution curves approach $y = -2$ on both sides, but move away from $y = -1$ on both sides. Therefore, $y = -2$ is a stable equilibrium solution, while $y = -1$ is an unstable equilibrium solution.

Keep in mind that we can also classify equilibrium solutions without sketching a direction field, by using the function $f(y)$ to determine the direction of the solution curves. For instance, to continue with the last example, once we've found the equilibrium solutions $y = -1,\ -2$, we'll pick a test-value in each of the three intervals they create, $y < -2$, $-2 < y < -1$, and $-1 < y$. We'll use test-values $y = -3$, $y = -5/4$, and $y = 0$.

We'll substitute our test values into $f(y) = y^2 + 3y + 2$.

$$f(-3) = (-3)^2 + 3(-3) + 2 = 9 - 9 + 2 = 2 > 0$$

$$f\left(-\frac{5}{4}\right) = \left(-\frac{5}{4}\right)^2 + 3\left(-\frac{5}{4}\right) + 2 = \frac{25}{16} - \frac{15}{4} + 2 = -\frac{3}{16} < 0$$

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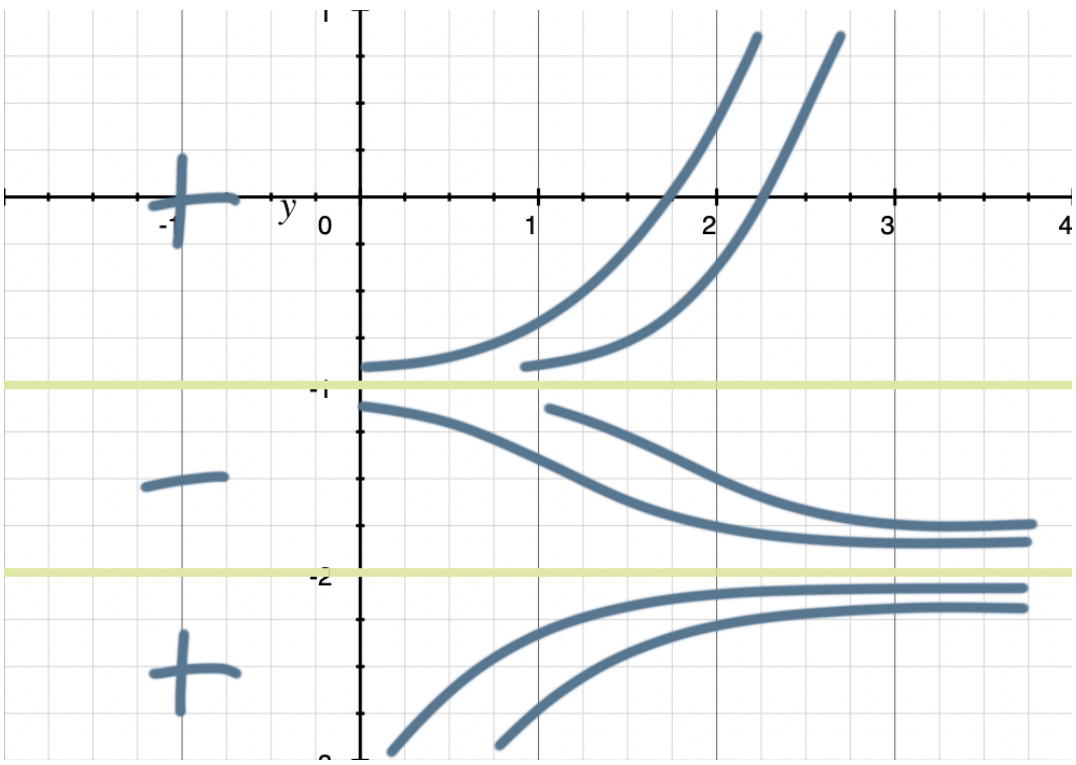
$$f(0) = 0^2 + 3(0) + 2 = 2 > 0$$

We find a positive value when $y = -3$, a negative value when $y = -5/4$, and a positive value when $y = 0$. A positive outcome from this test tells us that the solution curves are increasing, or rising in the interval, while a negative outcome from this test tells us that the solution curves in that interval are decreasing, or falling.

Let's summarize the results for this example in a table,

Interval	**Sign of** $f(y)$	**Direction of** $f(y)$
$(-1,\infty)$	$+$	Increasing/Rising
$(-2,-1)$	$-$	Decreasing/Falling
$(-\infty,-2)$	$+$	Increasing/Rising

and then plot these signs in the plane, without the direction field.



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With the signs plotted in between the equilibrium solutions, we can quickly see that the arrows are pointing toward and the solution curves will be approaching $y = -2$, while the arrows are pointing away from and the solution curves will be moving away from $y = -1$.

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